

Questions To Do

1. Check if a number is prime using a loop.

Input: A positive integer.

Output: "Prime" or "Not Prime".

2. Print the factorial of a number using a while loop.

Input: 5

Output: 120

3. Find the sum of digits of a number using a loop.

Input: 1234

Output: 10

4. Check if a number is a palindrome using a loop.

Input: 121 → *Output:* Palindrome

Input: 123 → *Output:* Not Palindrome

5. Find the least common multiple (LCM) of two numbers using a loop.

Input: 4, 6

Output: 12

6. Print a square matrix of numbers with diagonals as 0:

1 2 3 4

5 0 0 8

9 0 0 12

13 14 15 16

Hint : 1 Use **nested for loops**: one for rows (i) and one for columns (j).

2 .Maintain a **counter variable** (e.g., num) to print increasing numbers from 1.

3. Set the value to 0 **only when** the current element is on the **main diagonal** ($i == j$) or **anti-diagonal** ($i + j == n - 1$).

4. Otherwise, print the current number and increment it..

7. Continuous Numbers in Triangle:

Input: 10

Output:

1

2 3

4 5 6

7 8 9 10